

# AI/ML Internship Task 5 Report

Ensemble Learning, Model Optimization & Real-World ML Pipeline

## 1. Objective

The objective of this task was to implement an end-to-end machine learning workflow using ensemble learning techniques. The work focused on model development, optimization, comparison, and selection of the best performing classification model.

## 2. Dataset Description

The Breast Cancer Wisconsin dataset from Scikit-learn was used. It contains numerical features describing breast cancer samples with two target classes: Malignant and Benign.

## 3. Data Preprocessing

The dataset was divided into training and testing sets using stratified train-test split. Feature scaling was applied using StandardScaler for Logistic Regression.

## 4. Models Implemented

Three models were implemented:

- Logistic Regression as the baseline model.
- Random Forest Classifier as an ensemble learning model.
- Gradient Boosting Classifier as an advanced boosting technique.

## 5. Hyperparameter Tuning

Random Forest hyperparameters were optimized using GridSearchCV. Different parameter combinations were evaluated using cross-validation to improve model performance.

## 6. Cross Validation

K-fold cross-validation was performed to analyze model stability and ensure reliable performance across different data splits.

## 7. Feature Importance Analysis

Feature importance analysis was performed using Random Forest to understand which features contributed most to model predictions.

## 8. Model Performance Comparison

Models were evaluated using Accuracy, Precision, Recall, and F1 Score.

Final Results:

Logistic Regression:

Accuracy: 0.982456

Precision: 0.986111  
Recall: 0.986111  
F1 Score: 0.986111

## **9. Final Model Selection**

Based on evaluation metrics, Logistic Regression was selected as the final performing model.

Final Performance:  
Accuracy: 98.24%  
Precision: 98.61%  
Recall: 98.61%  
F1 Score: 98.61%

## **10. Conclusion**

This task provided practical experience with ensemble learning, model optimization, hyperparameter tuning, cross-validation, feature interpretation, and machine learning model comparison.